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PREDICTING WHERE PEOPLE LOOK AT ADS

A review of commercial and open attention-prediction models, and an open benchmark against human eye tracking

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ABSTRACT

Predicted-attention tools promise eye-tracking insight on any creative in seconds, and are now common in advertising pre-testing. We review 15 commercial predictive tools and five measured-attention vendors from public sources, and find that their headline accuracy claims ("92–95%") are mostly AUC expressed as a share of a human ceiling, a saturated metric on which almost every model scores highly. Only four commercial tools publish third-party-computed scores on the MIT/Tübingen benchmark. We then review the open-source state of the art, including licences, which rule out several leading models for commercial use, and benchmark six openly available models against human eye tracking on 149 webpages (FiWI), none of which they were trained on. DeepGaze MSDB is the most accurate (CC 0.676). UNISAL, the strongest model with a permissive licence, reaches CC 0.616 at 17× the speed of DeepGaze IIE. All models are weakest on text-heavy layouts. We release the benchmark and MediaAttention, an open toolkit that uses the benchmark's choice of model, so every number here can be reproduced.

1. INTRODUCTION

Attention is the first gate in advertising. A brand, product or message that is not looked at cannot persuade or be remembered, and eye-tracking studies of print ads show that pictures, text size and brand placement shape where attention goes and how it transfers between elements [17]. Visual salience alone can bias quick consumer choices, even against stated preferences [19]. Measuring attention with eye tracking is accurate but slow and costly, so a market has grown around *predicting* it: upload a creative, receive a heatmap and a set of scores.

For buyers, the difficulty is that these tools are hard to compare. Vendors quote accuracy figures computed on different metrics, test sets and viewing durations, often on their own data. This report makes four contributions:

1. **A market review** of 15 commercial predictive-attention tools and five measured-attention vendors, decoding what their accuracy claims mean (Section 3).
2. **An open-model review** covering accuracy, training data (for contamination) and licences (Section 4).

3. **An open benchmark** of six models against human eye tracking on data none of them was trained on (Section 5).
4. **MediaAttention**, an open-source toolkit that turns attention maps into ad metrics. Its default model is the benchmark's best permissively licensed model.

2. BACKGROUND

2.1 Attention-prediction models

Computational saliency began with biologically inspired feature models that combine intensity, colour and orientation contrast [14]. Since 2015, deep networks trained on large fixation datasets such as SALICON [4] (mouse-contingent viewing on 10,000 MS-COCO images) and MIT1003 [5] (eye tracking, 15 viewers) have dominated. The MIT/Tübingen benchmark [2, 29] evaluates models on held-out data (MIT300, CAT2000) and is the closest thing the field has to a neutral scoreboard.

2.2 Metrics

No single metric captures prediction quality [6]. We report six, all computed per image and averaged:

METRIC	COMPARES	READING
AUC	Prediction at fixated pixels vs. all pixels	Ranking quality; saturated (centre bias alone ≈ 0.78 on MIT300)
sAUC	Prediction at fixations vs. other images' fixations	Cancels centre bias; 0.5 = no better than a centre prior
NSS	Normalised prediction at fixations	Higher is better; 0 = chance
CC	Pearson correlation with the blurred fixation map	Our headline metric; 1 = identical maps
SIM	Histogram intersection of the two distributions	Overlap; 1 = identical
KLD	KL divergence of prediction from ground truth	Lower is better; punishes missed fixations

2.3 Datasets and contamination

Most models are trained on SALICON and MIT1003, natural photographs viewed freely for a few seconds. Advertising differs: it is dense with text, logos and layout. Design-oriented eye-tracking sets are rare. FiWI [26] covers 149 webpages (pictorial, text and mixed), with 11 viewers and 5 s free viewing. UEyes [27] covers 1,980 posters, webpages and app screens, with 62 participants and up to 7 s viewing. Because neither is in the training data of the models we test, both measure generalisation rather than memorisation.

3. COMMERCIAL TOOLS

We surveyed vendor websites, documentation, white papers and third-party listings in October 2026. Figures below are **as stated by each vendor** unless marked otherwise. "Not disclosed" means we found no public statement.

TOOL	APPROACH (AS DISCLOSED)	ACCURACY CLAIM	INDEPENDENT EVIDENCE	INPUTS · API · PRICE
Neurons AI	CNN encoder–decoder trained on proprietary eye tracking (~20k participants); separate emotion, memory and cognitive-demand models [30]	"95%+"; tech paper: CC 0.944 on own ad data	None found; not on MIT300	Image, video · API · quote
Dragonfly AI	Patented biologically inspired algorithm (QMUL); no training data disclosed	"Within 10% of a human sample"; metric not given	None found	Image, video, web, shelf · API · quote
Attention Insight	CNN; 5.5M fixations	92.5% = MIT300 AUC 0.864 ÷ ceiling 0.934	**On MIT300:** CC 0.758	Image, URL, video · API · €29–299/mo
3M VAS	Feature-based model (edges, intensity, colour, faces)	"92%"; own study: ~85% of the ROC ceiling	Own white paper only	Image, video · no API · ~$550/yr
EyeQuant	"7 distinct neural models"; 1.6M data points	"90%+", AUC relative to the human ceiling	None current	Image, page · API (enterprise) · from $97/mo
expoze.io (alpha.one)	Proprietary neural net	"95%" = 0.87 vs 0.92 (likely AUC)	One peer-reviewed comparison [31]	Image, video · API · quote
Brainsight	Proprietary and MIT/Tübingen data	"94%"; metric not given	None found	Image, video, URL · API · €199–899/mo
Feng-GUI	Simulates a 5 s, 40-person test	"92% MIT-validated"	Not on current MIT300	Image, web, video · API · quote
Clueify	Model trained on eye tracking	"92% consistent"	**On MIT300:** AUC 0.881, CC 0.575	Image · quote
Emotiva	Not disclosed	Not disclosed	**On MIT300:** CC 0.743	Image, video · API · demo
MedullaAI	Not disclosed	Not disclosed	**On MIT300:** CC 0.670	Not disclosed
EyeSee predictive	11.5M gaze points	None	None	Packaging, visuals
Kantar LINK AI	Kantar eye-tracking database	None	None	Video
Amplified AttentionAI	Attention-decay prediction from 50B+ data points	None	None	Creative · API
VisualEyes / Loceye	Acquired by Neurons in 2022; discontinued	84–86% (CC, own data)	–	–

Table 1. Commercial predictive-attention tools (public sources, October 2026). Measured-attention vendors, which use panels with webcam or hardware eye tracking (Realeyes/Adverteyes, Amplified Intelligence, Lumen, Adelaide, Tobii/ATTEX), are a different product: they measure real viewers and model media-level attention, not a gaze map for one creative.

3.1 What "X% accurate" means

Where a claim can be decoded, it is usually **AUC divided by the inter-observer ceiling**. For example, Attention Insight's 92.5% is its MIT300 AUC (0.864) over the gold standard (0.934). AUC is saturated: on MIT300 every listed model scores between 0.84 and 0.89, and a centre-only prior already reaches

0.78. Expressed as a share of the ceiling, small differences look large and weak models look strong. On CC, the same tools separate clearly. Clueify ranks well on AUC (0.881) yet has the lowest CC among them (0.575).

MODEL (MIT300)	AUC	SAUC	NSS	CC	KLD	TYPE
DeepGaze MSDB	0.894	0.816	2.74	0.883	0.254	open, research
DeepGaze IIE	0.883	0.794	2.53	0.824	0.347	open, research
TranSalNet-Dense	0.873	0.747	2.41	0.807	1.014	open (MIT)
UNISAL	0.877	0.784	2.37	0.785	0.415	open (Apache-2.0)
Attention Insight	0.864	0.745	2.18	0.758	0.887	commercial
Emotiva	0.864	0.770	2.20	0.743	0.496	commercial
MedullEyes-V1	0.841	0.724	1.89	0.670	0.659	commercial
Clueify	0.881	0.765	1.49	0.575	0.889	commercial
Centre bias	0.783	–	1.10	0.446	0.951	baseline
Gold standard (humans)	0.934	0.883	3.14	0.983	0.060	ceiling

Table 2. MIT/Tübingen MIT300 leaderboard [29], accessed 21 Oct 2026. These are natural images with free viewing. Four commercial tools have third-party-computed entries, and all of them score below the best open models. The leaderboard computes a metric-specific map from probabilistic models, which favours entries that submit densities (DeepGaze, UNISAL).

3.2 In-domain claims are not benchmark results

The most detailed public technical document we found, Neurons' tech paper, reports CC 0.944 against held-out data from its own ad corpus and compares this with the best research model's MIT300 CC (0.883) [30]. The two numbers come from different test sets, ceilings and viewing durations (5 s vs ~3 s), so they cannot be compared directly. A fair comparison needs every model on the same held-out data, which is what Section 5 does for the open models. Secondary metrics (clarity, cognitive load, memory, emotion) are defined differently by each vendor, and we found no external validation of any of them.

4. OPEN MODELS

MODEL	YEAR	TRAINED ON	MIT300 CC	CODE / WEIGHTS LICENCE	LAPTOP-FRIENDLY
DeepGaze MSDB [28]	2025	MIT1003, CAT2000, COCO-Freeview, DAEMONS, FIGRIM	0.883	No licence file	Yes (slow)
DeepGaze IIE [1]	2021	SALICON → MIT1003	0.824	No licence file	Yes
TranSalNet [32]	2022	SALICON	0.807	MIT	Yes
UNISAL [25]	2020	SALICON, DHF1K, Hollywood-2, UCF-Sports	0.785	Apache-2.0	Yes (4M params)
MSI-Net	2020	SALICON	0.779	MIT	TensorFlow
SUM [33]	2024	SALICON, MIT1003, CAT2000, OSIE, SalECI, UEyes	–	MIT	CUDA only

MODEL	YEAR	TRAINED ON	MIT300 CC	CODE / WEIGHTS LICENCE	LAPTOP-FRIENDLY
UMSI	2020	Imp1k importance (clicks, not gaze)	–	Non-commercial	Keras

Table 3. Open image models. Licences matter: the most accurate family (DeepGaze) ships without a licence file, which by default means all rights reserved. Video models (ViNet, TMFI-Net, SalFoM) score well on DHF1K. On real TV ads, a recent study found per-frame DeepGaze IIE reached CC 0.231 against 0.560 for the video model ViNet-S [34], so image models applied frame by frame understate attention dynamics in video.

5. BENCHMARK

5.1 Setup

- **Data.** FiWI [26]: 149 webpages at 1360×768, 11 viewers, 5 s free viewing, EyeLink 1000. Ground truth is all viewers' fixations, blurred with the dataset's own Gaussian (σ = 32 px) for CC, SIM and KLD. None of the six models was trained on FiWI.
- **Models.** We ran each model as released, without fine-tuning:
  - DeepGaze IIE and DeepGaze MSDB, with a Gaussian centre-bias prior; MSDB uses averaged dataset parameters and the dataset's 50 px per degree.
  - TranSalNet-Dense, SALICON weights, with its own resize and padding.
  - UNISAL, SALICON weights, as packaged in MediaAttention (identical to the original repository's output, CC > 0.9999).
  - Two baselines: a Gaussian centre prior and spectral residual [3].
- **Metrics.** pysaliency [2], with AUC computed over all pixels, sAUC over other images' fixations, and MIT/Tübingen KLD regularisation. All metrics are computed per image, then averaged.
- **Hardware.** Apple M3 Pro, PyTorch MPS. Timing is per-image inference after the model has loaded.

5.2 Results

MODEL	CC ↑	NSS ↑	SAUC ↑	AUC ↑	SIM ↑	KLD ↓	S / IMAGE	LICENCE
DeepGaze MSDB	0.676	1.545	0.748	0.840	0.642	0.462	2.90	no licence file
DeepGaze IIE	0.649	1.504	0.729	0.824	0.629	0.504	0.67	no licence file
UNISAL	0.616	1.306	0.661	0.796	0.613	0.575	0.04	Apache-2.0
TranSalNet-Dense	0.608	1.254	0.646	0.788	0.605	0.603	0.10	MIT
Centre-bias baseline	0.345	0.622	0.500	0.676	0.476	0.899	0.01	–
Spectral residual	0.209	0.403	0.562	0.633	0.449	1.052	0.00	–

Table 4. FiWI, 149 images. Mean per-image scores; ↑ higher is better, ↓ lower is better.

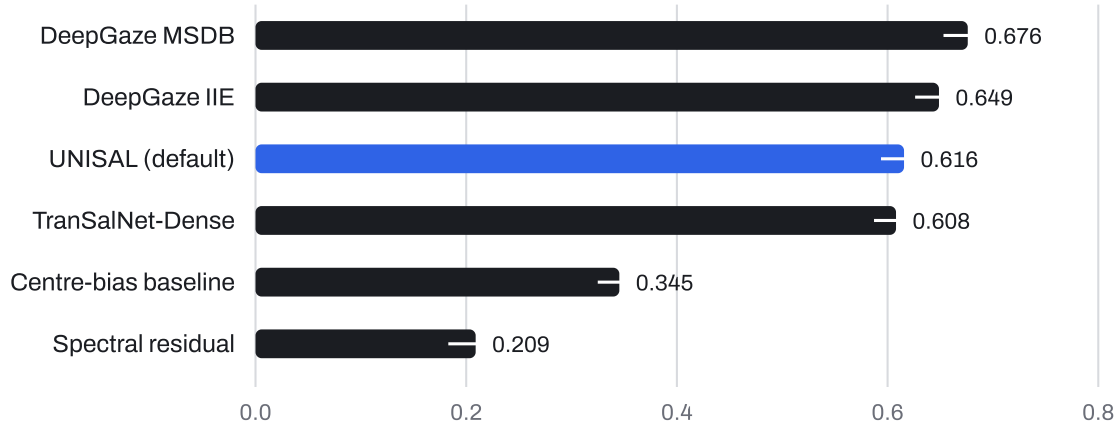


Figure 1. Correlation (CC) between each model's prediction and human fixations on FiWI. White ticks show 95% confidence intervals (±1.96 SE over images). The highlighted model is MediaAttention's default.

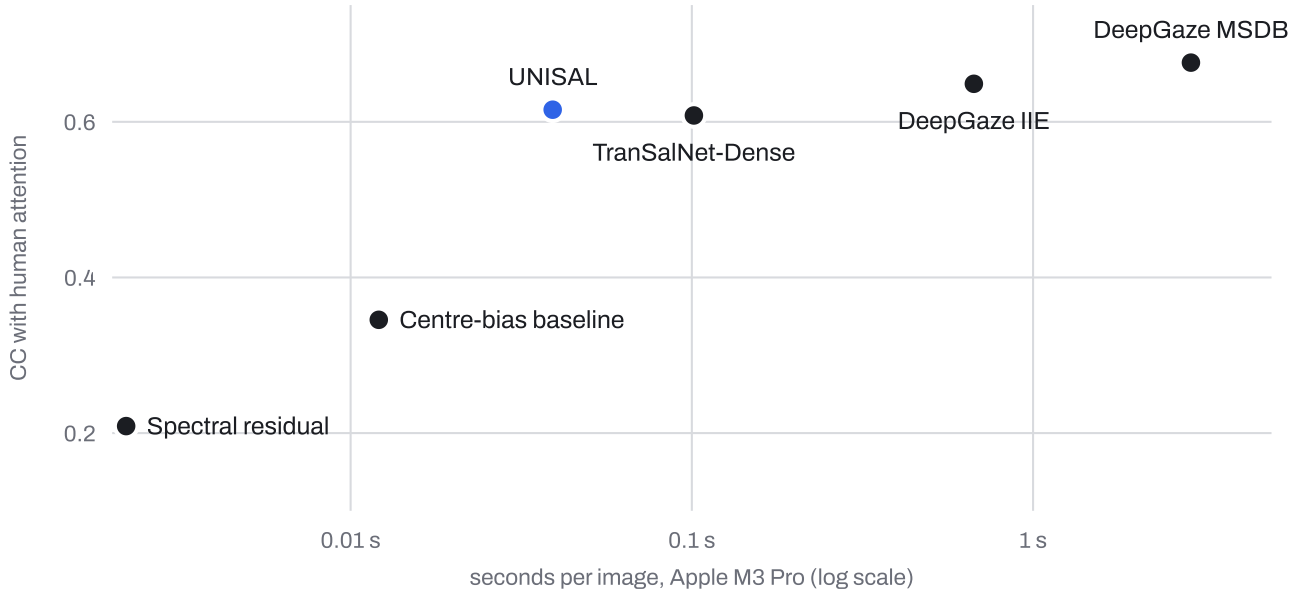


Figure 2. Accuracy against speed. DeepGaze MSDB is the most accurate. UNISAL is the fastest learned model and the most accurate one with a permissive licence.

5.3 Findings

> **Ranking.** DeepGaze MSDB leads on every metric (CC 0.676), followed by DeepGaze IIE (0.649), UNISAL (0.616) and TranSalNet (0.608). Every learned model beats the centre prior (CC 0.345) by a wide margin. Spectral residual, a classic low-level model, falls below the centre prior.

The licence–accuracy trade-off is small. UNISAL trails DeepGaze IIE by 0.033 CC and MSDB by 0.060. It runs at 0.04 s per image, about 17× faster than IIE, and is the only model in the top three that can be redistributed and used commercially without asking. That is why MediaAttention uses it by default and offers DeepGaze as an option.

Text is the hard case. Every model scores lowest on text-heavy pages: UNISAL scores 0.531 on text pages against 0.676 on pictorial ones, and MSDB 0.592 against 0.751. Natural-image training data teaches models to look at faces and objects, not at the reading patterns that dominate text layouts.

Advertising copy is exactly this kind of content, which makes ad-specific training data the most valuable next step.

MODEL	MIXED	PICTORIAL	TEXT
DeepGaze MSDB	0.686	0.751	0.592
DeepGaze IIE	0.680	0.726	0.541
UNISAL	0.641	0.676	0.531
TranSalNet-Dense	0.642	0.650	0.532
Centre-bias baseline	0.373	0.305	0.358
Spectral residual	0.249	0.220	0.159

Table 5. CC by FiWI page type.

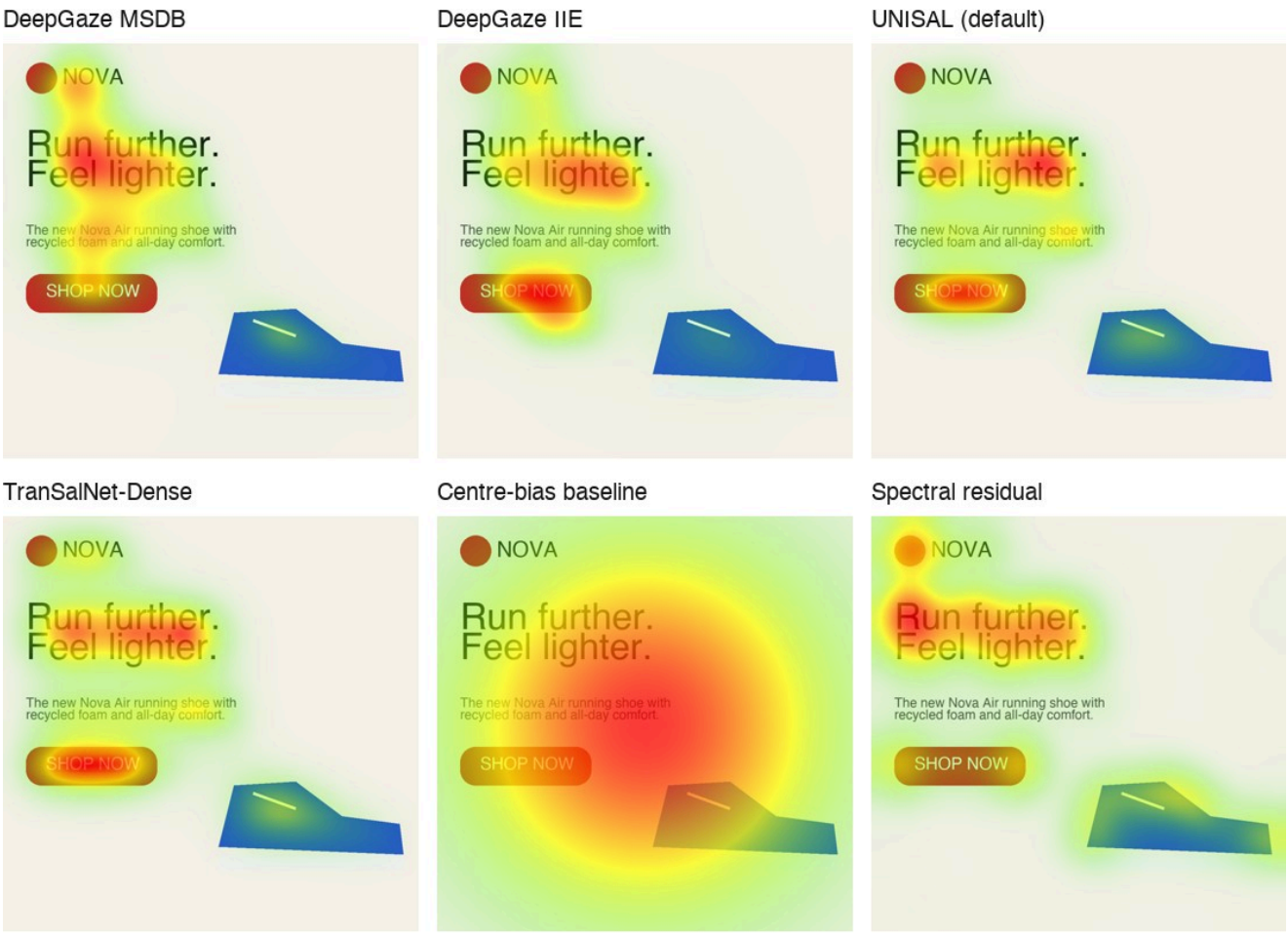


Figure 3. Each model's prediction on the same synthetic ad (MediaAttention's example). The learned models agree on the headline and call to action, and differ in how much attention they give the logo. Spectral residual fixates on high-contrast shapes.

Results on UEyes are added in the next revision of this report.

6. DISCUSSION

6.1 Questions to ask a vendor

1. Which metric? Ask for CC and NSS, not only AUC.
2. Which test set, was it held out, and was it scored by a third party?
3. Which viewing duration and which human ceiling?
4. How were secondary scores (clarity, memory, emotion) validated against real outcomes?

A tool that publishes MIT/Tübingen entries, or runs on an open benchmark such as this one, can be compared. A bare percentage cannot.

6.2 What predicted attention can and cannot tell you

Models trained on free viewing predict bottom-up, first-seconds attention: what pops out. They do not model goal-directed search (a shopper looking for a price), audio in video, or the placement context of an ad. Small independent comparisons of commercial tools with eye tracking point the same way: predicted maps track salience but miss goal-driven attention. Predicted attention is a fast first filter for comparing creative variants, not a replacement for testing with people when stakes are high.

7. LIMITATIONS AND NEXT STEPS

- **Size.** FiWI is small (149 images, 11 viewers) and covers webpages rather than ads. The next revision adds UEyes, which includes posters and ads.
- **Scope.** We evaluated only openly available models; commercial tools could not be run on the same data.
- **Video.** Video evaluation, an ad-tuned model trained on design eye-tracking data, and validation of secondary metrics against campaign outcomes are future work.

8. CONCLUSION

Commercial attention-prediction claims are rarely comparable, and where third-party scores exist, the best open models match or beat them. A permissively licensed open model, UNISAL, comes within a few hundredths of CC of the state of the art on unseen eye-tracking data while running in tens of milliseconds. The benchmark, the toolkit and this report are open, so these claims can be checked and extended.

APPENDIX A · REPRODUCIBILITY

```
git clone https://github.com/pranikz/MediaAttention && cd MediaAttention
pip install -e ".[unisal,deepgaze]" "git+https://github.com/matthias-k/pysaliency.git@dev"
pandas scikit-image tensorboardX fire
python benchmarks/run.py fiwi --models
center,spectral,unisal,transalnet,deepgaze_iie,deepgaze_msdb
python docs/research/build.py
```

Dataset and third-party model locations are documented in `benchmarks/run.py`. Per-image scores are in `benchmarks/results/`.

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Vendor information in Section 3 is from public websites, documentation and listings and may change. Vendor accuracy figures are their claims, not our measurements.